

```

226.     # Check if the player can move a token from home:
227.     if board[home] > 0 and board[track[flipTally]] == EMPTY:
228.         validMoves.append('home')
229.
230.     # Check which spaces have a token the player can move:
231.     for trackSpaceIndex, space in enumerate(track):
232.         if space == 'H' or space == 'G' or board[space] != player:
233.             continue
234.         nextTrackSpaceIndex = trackSpaceIndex + flipTally
235.         if nextTrackSpaceIndex >= len(track):
236.             # You must flip an exact number of moves onto the goal,
237.             # otherwise you can't move on the goal.
238.             continue
239.         else:
240.             nextBoardSpaceKey = track[nextTrackSpaceIndex]
241.             if nextBoardSpaceKey == 'G':
242.                 # This token can move off the board:
243.                 validMoves.append(space)
244.                 continue
245.             if board[nextBoardSpaceKey] in (EMPTY, opponent):
246.                 # If the next space is the protected middle space, you
247.                 # can only move there if it is empty:
248.                 if nextBoardSpaceKey == 'l' and board['l'] == opponent:
249.                     continue # Skip this move, the space is protected.
250.                 validMoves.append(space)
251.
252.     return validMoves
253.
254.
255. if __name__ == '__main__':
256.     main()

```

Exploring the Program

Try to find the answers to the following questions. Experiment with some modifications to the code and rerun the program to see what effect the changes have.

1. What happens if you change `nextTrackSpaceIndex == len(track) - 1` on line 152 to `nextTrackSpaceIndex == 1`?
2. What happens if you change `result = random.randint(0, 1)` on line 106 to `result = 1`?
3. What error do you cause if you change `board = {X_HOME: 7, X_GOAL: 0, O_HOME: 7, O_GOAL: 0}` on line 184 to `board = {}`?